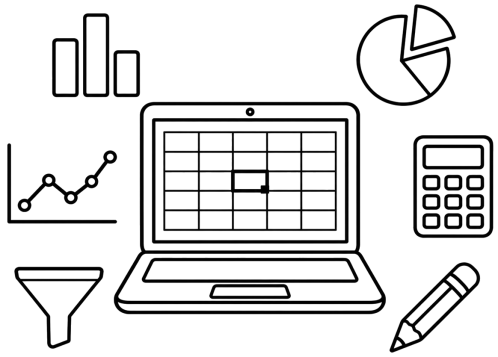
ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 10



UNIT 7
Electronic Spreadsheet

Student Workbook

"The King of ICT"

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STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Say what an electronic spreadsheet does and name spreadsheet software.
- [ ] Name the parts of the MS Excel 2010 and LibreOffice Calc windows.
- [ ] Use columns, rows, cell addresses and ranges of cells such as B2:C4.
- [ ] Enter labels, values and formulas, and work out formulas in order.
- [ ] Use SUM, AVERAGE, MAX, MIN, COUNT, COUNTA, ROUND, IF and RANK.
- [ ] Format cells and copy formulas with relative and absolute references.
- [ ] Sort and filter data, and create column, pie and XY scatter charts.

THREE RULES FOR THE COMPUTER ROOM

- **Always** save your work often, with your name in the file name.
- **Never** eat or drink near the computers.
- **Always** ask your teacher before you install or delete any software.

HOW TO USE THIS BOOK

Each chapter has three parts. First you **read** the chapter with your teacher. Then you **do** the practical task in a spreadsheet program. Last you **write** the exercises in this book.

You will need: a pencil, an eraser and a ruler, your textbook (pages 194 - 244), and a computer with MS Excel 2010 or LibreOffice Calc.

THE WORDS YOU WILL LEARN

Spreadsheet	Worksheet	Cell	Column
Row	Active cell	Cell address	Range of cells
Name Box	Formula Bar	Ribbon	Sheet tab
Label	Value	Formula	Operator
Function	Format Cells	Fill handle	Relative reference
Absolute reference	Sorting	Filtering	Chart

"A spreadsheet never gets tired of numbers. Learn it well, and it will work for you all your life."

- THE KING OF ICT

CHAPTER 1

The Spreadsheet and Its Window

? THE CLASS TRIP MONEY

"I added 42 amounts three times and got three different totals!"

A **spreadsheet** is like a square ruled book on the computer: rows, columns and cells. And it does the calculations for you, quickly and accurately!

7.1 Electronic spreadsheets

For systematic calculations we use a square ruled book. Electronic spreadsheets are based on this model of rows and columns.

i WITH A SPREADSHEET WE CAN DO

- Simple and complex calculations
- Presentation of data in charts
- Show data in ascending and descending order
- Segregate only the required data
- Check the validity of data
- Protection of data using passwords
- Saving for future use

Table 7.1 - spreadsheet software

Software	Company
Excel	Microsoft Corporation
Numbers	Apple Inc
LibreOffice Calc	The Document Foundation
OpenOffice Calc	Apache Foundation

MS Excel 2010: Start → Programs → Microsoft Office → Microsoft Office Excel 2010

LibreOffice Calc: Start → Programs → LibreOffice → LibreOffice Calc

7.2 The MS Excel 2010 window

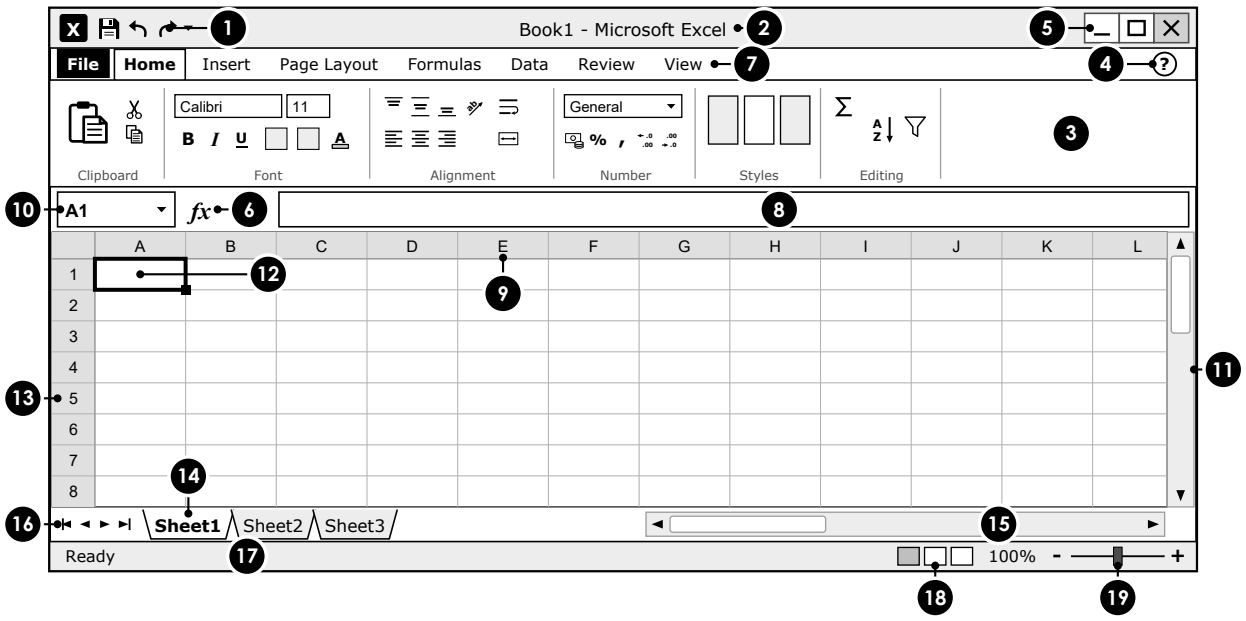


Figure 7.1 - the user interface of MS Excel 2010 (the numbers of textbook Table 7.2)

No	Part	What it does
1	**Quick Access Tool Bar**	Save, Undo, New, Print Preview
2	**Title Bar**	Name of the application software
3	**Ribbon**	Toolbar with Home, Insert ... tools
4	**Help**	Provides help functions
5	**Control buttons**	Minimize, open or expand the window
6	**Insert Function**	Opens a dialog box to make formulas
7	**Tabs**	Change the ribbon
8	**Formula Bar**	Shows the formula / cell contents
9	**Column Headings**	Show the column names
10	**Name Box**	Shows the active cell address

No	Part	What it does
11	**Vertical Scroll Bar**	Scrolls the worksheet vertically
12	**Active Cell**	The cell in which data is entered
13	**Row Headings**	Show the row numbers
14	**Sheet Tab**	Represents a worksheet
15	**Horizontal Scroll Bar**	Scrolls the worksheet horizontally
16	**Worksheet Navigation**	Changes the worksheet
17	**Status Bar**	Shows the status of the worksheet
18	**View Buttons**	Change the view of the worksheet
19	**Zoom Control**	Zooms the view in or out

7.3 The LibreOffice Calc window

The LibreOffice Calc 4.1 window (textbook Figure 7.2) has most of the same parts. Instead of the ribbon and the tabs, it has three bars:

Part	What it does
Menu Bar	Helps to select commands (File, Edit, View, Insert, Format ...).
Standard Tool Bar	Useful to select the standard tools (New, Open, Save, Print ...).
Formatting Tool Bar	Helps to format the worksheet (font, size, bold, alignment ...).

7.4 Columns, rows and the active cell

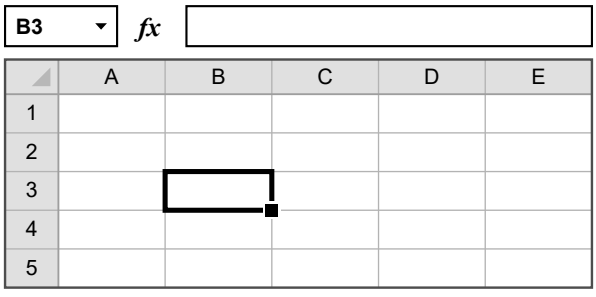


Figure 7.4 - B3 is the active cell

- A **worksheet** is made of cells formed by columns and rows. Each worksheet has a **sheet tab**.
- **Columns**: A, B ... Z, then AA ... AZ, BA ... BZ. **Rows**: 1, 2, 3 ...
- MS Excel 2010 has 1,048,576 (2^{20}) rows and 16,384 (2^{14}) columns.
- The selected cell is the **active cell**. It has a **thick border**.
- A **cell address** is the column letter, then the row number: **B3**. The **Name Box** shows it.

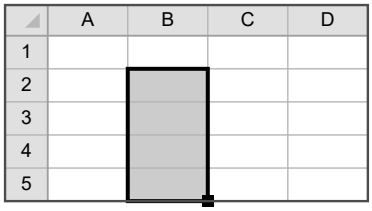
7.5 Moving around the worksheet

Key(s)	Moves the active cell ...
Arrow keys	one cell left, right, up or down
Ctrl + Arrow key	to the end of the data range
Home	to column A of the same row
Ctrl + Home	to cell A1

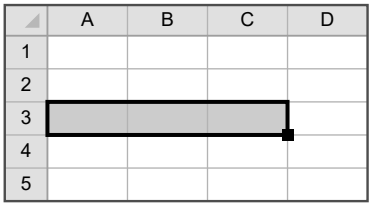
Key(s)	Moves the active cell ...
Ctrl + End	to the bottom right cell of the data
Page Up	one screen up
Page Down	one screen down

7.6 Ranges of cells

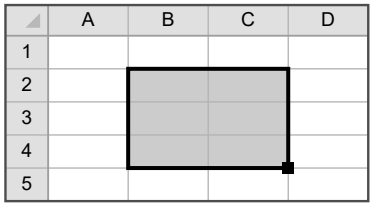
A block of adjacent cells that is selected is a **range of cells**: the first cell, a colon (:) and the last cell. Hold **Ctrl** to select more than one range.



B2:B5 - a range along a column: the column letter stays the same



A3:C3 - a range along a row: the row number stays the same



B2:C4 - several rows and columns: B2, B3, B4, C2, C3, C4

DO IT - PRACTICAL 1 (TEXTBOOK ACTIVITIES, PAGES 195 - 202)

Explore the spreadsheet window. **Ask your teacher first.**

1	Start MS Excel 2010 or LibreOffice Calc from the Start menu (see 7.1).
2	Find the parts of 7.2 in the window. Name each part to your partner.
3	Click D3 and look at the Name Box. Try Ctrl + Home, Page Down and Home.
4	Drag from B2 to C4 to select B2:C4. Hold Ctrl and select E2:E5 too.

A What does it do?

6 marks

Match the part with what it does. Write the letter.

A - Part		B - What it does
1. Name Box		**A.** Represents a worksheet.
2. Formula Bar		**B.** Opens a dialog box to create formulas.
3. Sheet Tab		**C.** Shows the address of the active cell.
4. Tabs		**D.** Displays the status of the worksheet.
5. Status Bar		**E.** Shows the formula and the contents of a cell.
6. Insert Function		**F.** Change the ribbon.

B Which key?

6 marks

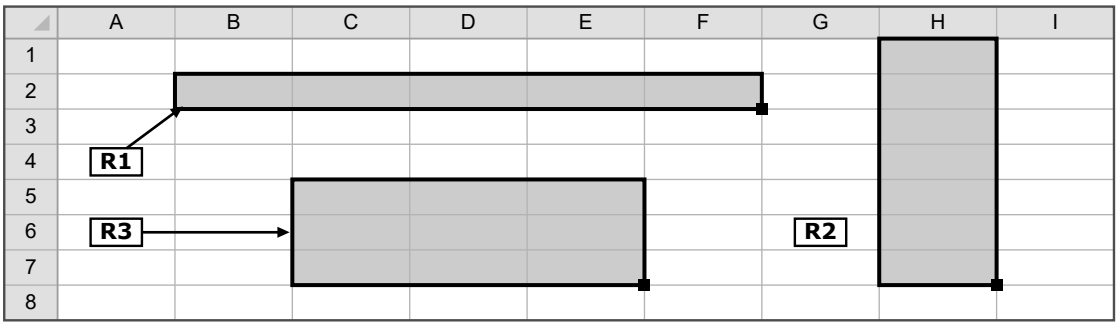
Write the key or keys for each job.

You want to ...	Key(s)
Go to cell A1.	
Move the active cell one cell down.	
Go to column A of the same row.	
Show the next screen of the worksheet below.	
Go to the bottom right cell of the data range.	
Go to the end of the data range on the right.	

C Ranges in a worksheet

8 marks

Textbook Activity, page 202. Look at R1, R2 and R3.



1. What is the address of the first cell of the range R1? .....
2. What is the address of the last cell of the range R1? .....
3. Write the cell range R1. .....
4. Write the cell range R2. .....
5. Which range (R1, R2 or R3) is along a row? .....
6. Which range (R1, R2 or R3) is along a column? .....
7. Write the cell range R3. .....
8. How many cells are there in R3? .....

D Fill in the blanks

8 marks

Use the words in the box.

WORD BANK

columns letter number active thick address range Ctrl

1. A worksheet is made up of cells formed by ..... and rows.
2. Each column is named with an English ..... or a combination of letters.
3. Each row of a worksheet is named with a ..... .
4. The cell that is selected now is called the ..... cell.
5. An active cell always has a ..... border.
6. The Name Box shows the cell ..... of the active cell.
7. A block of adjacent cells that is selected is called a ..... of cells.
8. To select more than one range of cells, use the ..... key.

E True or false

6 marks

Write T or F in the box.

1.		Numbers is a spreadsheet application made by Apple Inc.
2.		LibreOffice Calc is made by Microsoft Corporation.
3.		In a cell address, the row number is written first.
4.		The column after Z is named AA.
5.		In the range A3:C3 the row number stays the same.
6.		A worksheet of MS Excel 2010 has 16,384 rows.

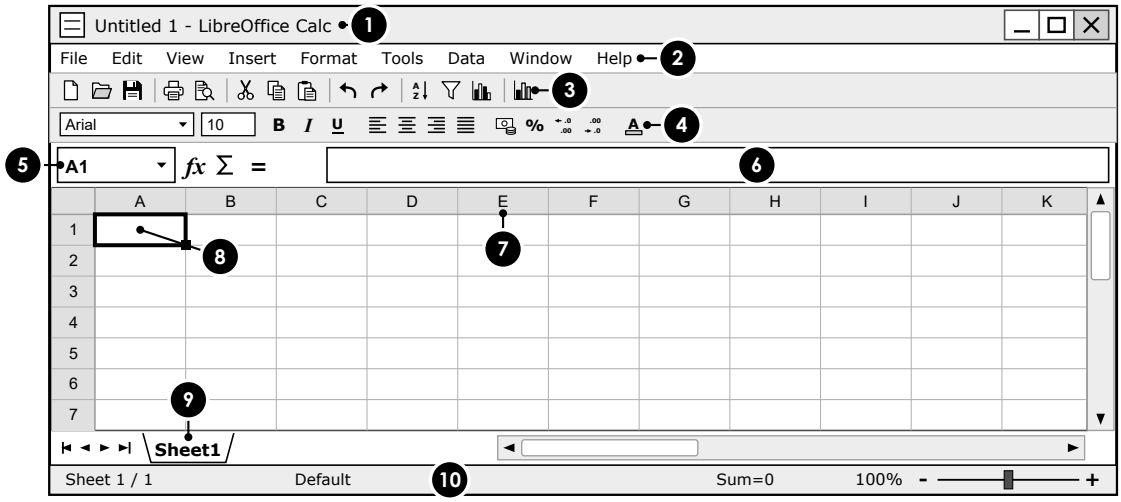
F Label the Calc window

10 marks

This is the LibreOffice Calc window. Write the name of each numbered part.

WORD BANK

Status Bar	Name Box	Menu Bar	Zoom Control	Active Cell	Title Bar
Sheet Tab	Formula Bar	Row Heading	Formatting Tool Bar	Column Heading	Standard Tool Bar



1. ..... 2. ..... 3. ..... 4. ..... 5. .....

6. ..... 7. ..... 8. ..... 9. ..... 10. .....

CHAPTER 2

Data, Operators and Formulas

? 14 OR 18?

"Nimal says that =2+4*3 gives 18. Kamala says 14. Who is right?"

Kamala! A spreadsheet does * before +. In this chapter: labels, values, formulas and the order of the **operators**.

7.7 Labels, values and formulas

Select a cell, then enter the data. The content of a cell is a label, a value or a formula:

Kind	What it is	Aligned
1. Label	Letters, numbers, special symbols or text typed with the keyboard.	left
2. Value	Data with a numerical figure: numbers, dates, times, money.	right
3. Formula	Begins with = and uses values, cell addresses and functions.	-

	A	B	C
1	**Name**	**Town**	**Telephone**
2	Piumi Kaushalya	Kegalle	0714365759
3	Hakeem Anver	Matara	0778746853

Figure 7.10 - labels are left aligned

C1 *fx* =A1+B1

	A	B	C	D
1	2	3	5	

Figure 7.12 - the cell shows the result; the Formula Bar shows the formula

	A	B
1	Integer	23
2	Decimal	5.72
3	Fractional	3 1/4
4	Percentage	65%
5	Scientific	4.37E+04
6	Date	5/12/2007
7	Time	2:45 PM
8	Currency	Rs 400.00

Figure 7.11 - values are right aligned

A telephone number typed as a value loses its first 0, so format those cells as **Text** first (7.17). For a fraction, type a space after the whole number: **3 1/4**.

7.8 Mathematical operators and their order

Operator	Meaning
+	addition
-	subtraction
*	multiplication
/	division
^	exponentiation (power)

Operator precedence

Order	Operators
First	**( )**
Second	**^**
Third	*** /**
Fourth	**+ -**

Table 7.4

Formula	Result
=2+4	6
=5-2	3
=4*5	20
=12/6	2
=2^4	16

Brackets change the order: the expressions within brackets are evaluated first. Operators of the same level (* and /, or + and -) are done from **left to right**.

Table 7.5 - brackets change the result

Formula	Evaluation	Result
=2+4*3	=2+12	**14**
=(2+4)*3	=6*3	**18**
=4*3^2	=4*9	**36**
=(4*3)^2	=12^2	**144**

Table 7.6 - step by step

Steps	What is done
=5+2*3^2/6-3	the formula
=5+2*9/6-3	3^2 first (^)
=5+18/6-3	2*9 (left to right)
=5+3-3	18/6
=8-3	5+3 (left to right)
= 5	the answer

7.9 Formulas with cell addresses

When a formula uses **cell addresses**, its result is **updated** by itself when those values change. **Example 1** (Figure 7.13): the length of a rectangle is in A2 and the breadth in B2.

	A	B	C	D
1	**Length**	**Breadth**	**Perimeter**	**Area**
2	45	24	=A2+A2+B2+B2	=A2*B2
3			=A2*2+B2*2	
4			=(A2+B2)*2	

C2, C3 and C4 give the perimeter **138** in three ways. D2 gives the area **1080**.

Write nothing before the = sign: **D2=A2*B2** is only a label!

> "First the brackets, then the powers. Order matters - in formulas and in life."
>
> - THE KING OF ICT

	A	B
1	Price of a Book	48
2	Price of a Pencil	12
3	Price of book and pencil	=B1+B2
4	Difference in price	=B1-B2
5	Price of 4 books	=B1*4
6	Price of 4 books and 2 pencils	=B1*4+B2*2

Example 2 *(Figure 7.16): B3 to B6 show 60, 36, 192 and 216*

	A	B
1	Side length of the cube	8
2	Side area of the cube	=B1^2
3	Total area of the cube	=6*B1^2
4	Volume of the cube	=B1^3

Example 3 *(Figure 7.17): B2 to B4 show 64, 384 and 512*

DO IT - PRACTICAL 2 (TEXTBOOK ACTIVITIES, PAGES 206 - 208)

Formulas with cell addresses. **Ask your teacher first.**

1	Make the rectangle worksheet of Figure 7.13 with its four formulas.
2	Type new values in A2 and B2. Watch C2, C3, C4 and D2 change.
3	Make Figures 7.16 and 7.17. Change B1 and B2 and watch the results.
4	Type **D2=A2*B2** in an empty cell: it stays a label! Save your work.

A Work them out

10 marks

Textbook Activity, page 206. Write the result of each formula.

1. **=6-4/2** .....
2. **=5*4+3** .....
3. **=2^5+3** .....
4. **=4+5*2^3** .....
5. **=8/4*2** .....
6. **=1+2^2*3** .....
7. **=5/2+1^3*4** .....
8. **=3-4*3/2** .....
9. **=3*(2+4)/9+1** .....
10. **=2^3/(5-1^3)*5** .....

B Results from a worksheet

12 marks

Textbook Activity, page 208 (Table 7.7). Use the order of the operators.

	A	B	C	D	E
1	7	5	4	2	
2	3	1	9	4	
3	6	2	4	8	
4					

Figure 7.18

Look at the worksheet. **=A1+B1** gives 12, **=C1+C2+C3** gives 17, **=B2+C3+B3** gives 7 ... An empty cell counts as 0.

Formula	Result
=A1*B1-D2	
=D1-C2/A2	
=B3^C3-A3	
=A1*C1/C3	
=D2*C4*E1	
=A3/B3+A2*D1	

Formula	Result
=A1+(B2*C2)-B3+C2/B2	
=B2-B3+B1*(D1+D2^B1)	
=A2^B3-C2+A3*(B3+C1)	
=A3/C3*(D2*D1)	
=D3/C3/B3*A3+B2	
=(D3/(A1+B2))^C2	

C Label, value or formula?

8 marks

Tick one box for each cell content.

Cell content	Label	Value	Formula
Piumi Kaushalya	☐	☐	☐
5.72	☐	☐	☐
=A1+B1	☐	☐	☐
Kegalle	☐	☐	☐
65%	☐	☐	☐
=B1^3	☐	☐	☐
5/12/2007	☐	☐	☐
D2=A2*B2	☐	☐	☐

D Step by step

6 marks

Textbook Table 7.6. Show every step, like 7.8, and the answer.

1. =8/2*3-2^3+5

2. =5+2*3^2/(6-3)

E Write the formulas

8 marks

The school canteen. Use cell addresses. Write the result too.

	A	B
1	Price of a bun	60
2	Price of a cup of tea	40

The price of a bun is in B1 and the price of a cup of tea is in B2. Write a formula for each cell. 1 mark for the formula, 1 mark for the result.

1. B3: the price of a bun and a cup of tea = ... Result: .....

2. B4: how much more a bun costs than a cup of tea = ... Result: .....

3. B5: the price of 5 buns = ... Result: .....

4. B6: the price of 3 buns and 4 cups of tea = ... Result: .....

F Spot the mistakes

4 marks

Kasun made FOUR mistakes. Underline them. Write the correct words below.

> Kasun wrote: "A formula always starts with a plus sign. Labels are right aligned by default. In =2+4*3 the addition is done first, so the answer is 18."

1. ..... 2. ..... 3. ..... 4. .....

CHAPTER 3

Functions and Formatting

? FIVE HUNDRED CELLS

"Sir, must I type =A1+A2+A3 ... all the way to A500?"

No! Type **=SUM(A1:A500)**. A **function** does a long calculation with a few letters. Then **format** the worksheet so that it is easy to read.

7.10 Functions

With a lot of data, formulas with cell addresses become long: **=A1+A2+A3+A4+A5+A6+A7+A8+A9+A10** gives the same result as **=SUM(A1:A10)**. A function needs only the first and the last cell address of the range (Table 7.8):

Function	Syntax	Use
SUM	=SUM(Number1, Number2 ...)	Adds the values in a range of cells.
AVERAGE	=AVERAGE(Number1, Number2 ...)	Finds the average of the values in a range.
MAX	=MAX(Number1, Number2 ...)	Finds the largest value in a range of cells.
MIN	=MIN(Number1, Number2 ...)	Finds the smallest value in a range of cells.
COUNT	=COUNT(Number1, Number2 ...)	Counts the cells that contain numerical values.
COUNTA	=COUNTA(Number1, Number2 ...)	Counts the non-empty cells in a range.
ROUND	=ROUND(Number, Num_digits)	Rounds a value to the required decimal places.
IF	=IF(Condition, Value_if_true, Value_if_false)	Checks the condition: true gives the first value, false gives the second.

7.11 SUM, AVERAGE, MAX and MIN

	A	B	C	D	E	F
1	2	6	4			
2	5	3	7		Student	
3	5	9	12		2	4.5847

Figure 7.19 - the worksheet for 7.11 and 7.12

A **colon** (:) gives a range: A1:C1 is A1, B1 and C1. A **comma** (,) separates cells and ranges.

Formula	It finds	Result
=SUM(A1,B1,C1)	the sum of A1, B1 and C1	**12**
=SUM(A1:C1,B2)	the sum of A1 to C1, and B2	**15**
=SUM(B1:C2)	the sum of the range B1 to C2	**20**
=SUM(A1:A3,C1:C3)	the sum of A1 to A3 and C1 to C3	**35**
=AVERAGE(A1:C1)	the average of A1 to C1	**4**
=AVERAGE(A1:C1,B2)	the average of A1 to C1, and B2	**3.75**
=MAX(A2:C2,B3)	the largest of A2 to C2, and B3	**9**
=MAX(A1,B1:C2)	the largest of A1, and B1 to C2	**7**
=MIN(A1,B2,C1)	the smallest of A1, B2 and C1	**2**
=MIN(A2:C2,B3)	the smallest of A2 to C2, and B3	**3**

7.12 COUNT and COUNTA

COUNT counts the cells that contain **numbers**. **COUNTA** counts the cells that are **not empty** (numbers and labels). In Figure 7.19, E2 has the label Student:

Formula	It counts	Result
=COUNT(A1:C1,B2)	cells with numbers in A1 to C1, and B2	**4**
=COUNT(B1:C3)	cells with numbers in B1 to C3	**6**
=COUNTA(A1:E3)	non-empty cells in A1 to E3	**11**
=COUNTA(A2:F2)	non-empty cells in A2 to F2	**4**

7.13 The ROUND function

=ROUND(Number, Num_digits) rounds a value to the required number of decimal places. A **negative** Num_digits rounds to the nearest 10 (-1) or 100 (-2): if the digit on its right is less than 5, round down; if it is 5 or more, round up. (F3 is 4.5847.)

Formula	Result
=ROUND(2.15,1)	**2.2**
=ROUND(2.149,1)	**2.1**
=ROUND(-1.475,2)	**-1.48**
=ROUND(21.5,0)	**22**

Formula	Result
=ROUND(21.5,-1)	**20**
=ROUND(123,-1)	**120**
=ROUND(456,-2)	**500**
=ROUND(F3,1)	**4.6**

7.14 The IF function

=IF(logical_test, value_if_true, value_if_false) checks a condition such as B2>10. If it is TRUE, it returns the first value; if it is FALSE, the second one. Commas separate the three parts.

	A	B	C	D	E
1	6	7			
2		12			
3			yes		
4				30	
5					100

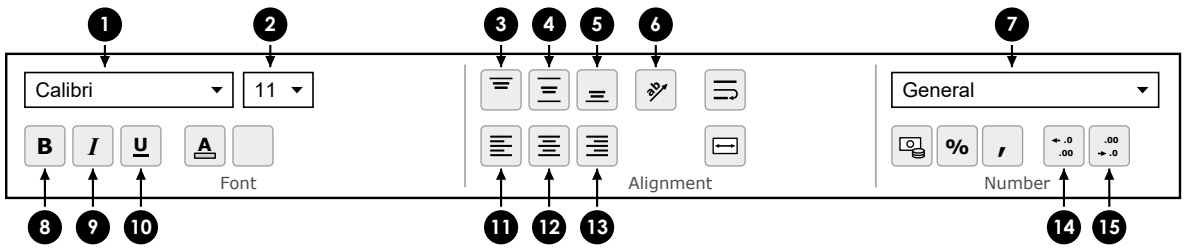
Figure 7.20

Words such as "Pass" are written in **double quotation marks**. Numbers, cell addresses and calculations (50, E5*1.1) are not.

Formula	Result
=IF(B2>10,"Pass","Fail")	**Pass**
=IF(C3="Yes",50,0)	**50**
=IF(A1="","Empty","Not Empty")	**Not Empty**
=IF(D4<=25,"Low","High")	**High**
=IF(E5>=100,E5*1.1,E5*1.2)	**110**
=IF(C2="Complete","","Pending")	**Pending**

7.15 Formatting tools

In MS Excel 2010 the formatting tools are on the **Home** tab of the ribbon (Figure 7.21, Table 7.17):

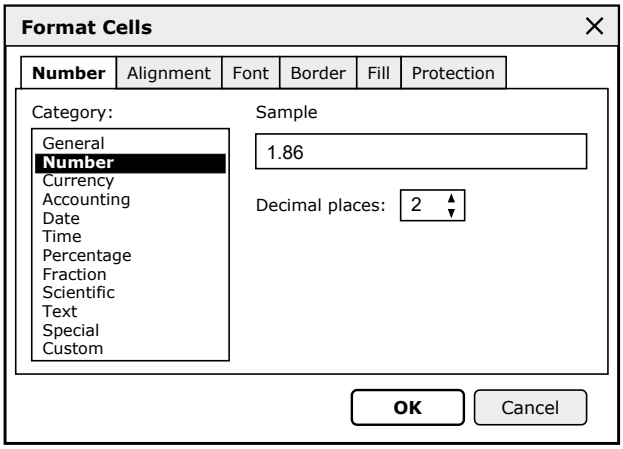


No	Tool	What it does
1	**Font**	Font types
2	**Font Size**	Font sizes
3	**Vertical Top Alignment**	Text to the top of the cell
4	**Vertical Middle Alignment**	Text to the middle of the cell
5	**Vertical Bottom Alignment**	Text to the bottom of the cell
6	**Orientation**	Rotates the text
7	**Number Format**	How values are shown
8	**Bold**	Bold text

No	Tool	What it does
9	**Italic**	Italic text
10	**Underline**	Underlined text
11	**Left Alignment**	Text to the left
12	**Center Alignment**	Text to the centre
13	**Right Alignment**	Text to the right
14	**Increase decimal places**	More decimal places
15	**Decrease decimal places**	Fewer decimal places

The **Formatting Tool Bar** of LibreOffice Calc has ten of these tools: Font, Font Size, Bold, Italic, Underline, Left, Center and Right Alignment, and Increase and Decrease decimal places (Table 7.18).

7.16 The Format Cells dialog box



Open it in MS Excel 2010 with **Home tab** → **Number group** (the small arrow of the group), and in LibreOffice Calc with **Format** → **Cells**.

1	Select the range of cells.
2	Open the **Format Cells** dialog box.
3	Click the **Number** tab.
4	Select the category (Number, Text, Percentage ...).
5	Change the decimal places, or choose the format you need.
6	Click **OK**.

7.17 Number formats and alignment

In the **Number** tab, choose a category (textbook 7.5.4 - 7.5.10). The value stays the same; only the way it is shown changes:

Category	Use it for	Value	Shown as
Number	a fixed number of decimal places	1.857142857	**1.86**
Text	numbers that must be labels (telephone numbers)	714365759	**0714365759**
Percentage	a quotient as a percentage	0.3662182362	**36.62%**
Scientific	very large or very small numbers	8796093022208	**8.80E+12**
Date	dates: 5/12/2014 = 12 May 2014 (mm/dd/yyyy)	5/12/2014	**12-May-14**
Time	times: hh:mm:ss AM/PM	10:35:53 AM	**10:35**
Currency	money, with Rs or $ in front	400	**Rs 400.00**

"Why add a hundred cells by hand? One function does it. Work smart, not only hard."

- THE KING OF ICT

	A	B	C
1	**Month**	**Income**	**As Percentage**
2	January	Rs. 24,500.00	36.62%
3	February	Rs. 14,000.00	20.93%
4	March	Rs. 21,600.00	32.29%
5	April	Rs. 6,800.00	10.16%

Table 7.19 - the school book shop: currency and percentage

Use the **Alignment** tab of Format Cells (or the alignment tools) to align labels and values to the **left**, **centre** or **right** of the cells.

DO IT - PRACTICAL 3 (TEXTBOOK ACTIVITIES, PAGES 210 - 225)

Functions and formatting. **Ask your teacher first.**

1	Make Figure 7.19 and type the formulas of 7.11 and 7.12. Check them.
2	Try the ROUND and IF formulas of 7.13 and 7.14 (Figures 7.19, 7.20).
3	Make the school library worksheet (textbook page 224) with SUM and AVERAGE.
4	Format it with Center Alignment, Bold, font size 16 and 2 decimal places.

A Round the numbers

6 marks

Write the result of each ROUND formula.

1. **=ROUND(3.456,2)** .....
2. **=ROUND(7.5,0)** .....
3. **=ROUND(1234,-2)** .....
4. **=ROUND(85,-1)** .....
5. **=ROUND(-2.345,1)** .....
6. **=ROUND(0.125,2)** .....

B Find the results

8 marks

Use the worksheet of Figure 7.19 in 7.11.

Formula	Result
=SUM(A2:C2)	
=AVERAGE(A1:A3)	
=MAX(A1:C3)	
=MIN(B1:C3)	

Formula	Result
=COUNT(A1:F3)	
=COUNTA(D1:F3)	
=ROUND(F3,2)	
=SUM(A1:B2)+C3	

C Pass or fail?

6 marks

The IF function.

	A	B	C
1	**Name**	**Marks**	**Result**
2	Kavindu	45	
3	Fathima	72	
4	Tharushi	38	
5	Sajith	50	

C2 has **=IF(B2>=50,"Pass","Fail")**. C3, C4 and C5 have the same formula for their own row.

1. Write what C2, C3, C4 and C5 show. (4 marks)

2. D2 must show **Good** if the mark in B2 is more than 70, and **Try again** if not. Write the formula. (2 marks)

.....

D Name the tools

8 marks

Write the name of each formatting tool.

1. 
2. 
3. 
4. 
5. 
6. 
7. 
8. 

E ICT marks of three terms

10 marks

Textbook Activity, page 216. ab = absent. Use functions.

	A	B	C	D	E
1	**Student Name**	**Term 1**	**Term 2**	**Term 3**	
2	K.Bhanu Somarathna	57	79	66	
3	Wasantham Jayaraj	45	57	60	
4	Mohommad Niyas	89	ab	76	
5	R.Ganga Kumari	71	62	78	
6	P.Vasuki Devi	60	70	65	
7	M.M.Fawsiya	42	58	67	
8					

Write the formula with a function, and the result it shows. 1 mark each.

1. E2: the total marks of K. Bhanu Somarathna — Formula: ..... Result: .....
2. E3: the average marks of Wasantham Jayaraj — Formula: ..... Result: .....
3. B8: the highest mark in Term 1 — Formula: ..... Result: .....
4. C8: the number of students who sat Term 2 — Formula: ..... Result: .....
5. D8: the lowest mark in Term 3 — Formula: ..... Result: .....

F The school library

8 marks

Figure 7.31. Write the formula and the result (averages with 2 decimal places).

	A	B	C	D	E	F
1	Usage of School Library as Section wise					
2	**Section**	**January**	**February**	**March**	**Total**	**Average**
3	*Art*	14	20	16		
4	*Commerce*	10	13	14		
5	*Science*	18	20	17		
6	Total					
7	Average					

1. B6: the January total = ... Result: .....
2. B7: the January average = ... Result: .....
3. E3: the Art total = ... Result: .....
4. F3: the Art average = ... Result: .....

G Which format?

7 marks

Write the category of the Number tab you would use.

You want to ...	Category
show the telephone number 0714365759 with its first 0	
show 0.25 as 25%	
show 8796093022208 as 8.80E+12	
show 1.857142857 as 1.86	
show 400 as Rs 400.00	
show 10:35:53 AM as 10:35	
show 5/12/2014 as 12-May-14	

CHAPTER 4

Copying Formulas, Sorting and Charts

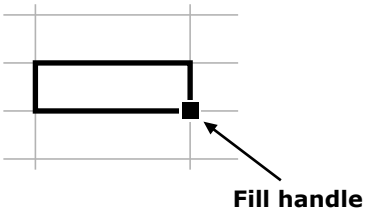
? ONE FORMULA, FORTY STUDENTS

"Must I type a total formula for every one of the forty students?"

No. Type it once and drag the **fill handle**; the cell addresses change by themselves. Then keep some fixed with **$**, rank, sort and filter the data, and draw **charts**.

7.18 Copying a formula

A class teacher types the total and average formulas once and copies them to every student. The small black square at the bottom-right corner of the selected cell or range is the **fill handle**.



	A	B	C
1			=A1+B1
2			
3			
4			

drag down

With the fill handle: select the cell with the formula, click the fill handle and drag it to the required cell.

With Copy and Paste: select the cell, copy it (**Ctrl + C**), select the new cell and paste (**Ctrl + V**).

7.19 Relative and absolute cell references

A cell address in a formula is a **relative reference**: it changes when the formula is copied. A dollar sign (**$**) before the row number or the column letter keeps that part **absolute** (fixed).

Address	Its name	When the formula is copied ...
H2	Relative cell reference	the column letter and the row number change
H$2	Row absolute cell reference	the row number stays; the column letter changes
$H2	Column absolute cell reference	the column letter stays; the row number changes
$H$2	Row and column absolute cell reference	nothing changes

Figures 7.33, 7.34, 7.36 and 7.39: a formula in C1 is dragged down column C and along row 1:

In C1	Copied to C2	to C3	to D1	to E1
=A1+B1	=A2+B2	=A3+B3	=B1+C1	=C1+D1
=A$1+B$1	=A$1+B$1	=A$1+B$1	=B$1+C$1	=C$1+D$1
=$A1+$B1	=$A2+$B2	=$A3+$B3	=$A1+$B1	=$A1+$B1
=$A$1+$B$1	=$A$1+$B$1	=$A$1+$B$1	=$A$1+$B$1	=$A$1+$B$1

7.20 Absolute references at work

	A	B	C
1	**Pencils**	**Price**	**Amount (Rs)**
2	1	10.00	10.00
3	2		20.00
4	3		30.00
5	4		40.00
6	5		50.00
7	6		60.00

Figure 7.35

The price of pencils. C2 has **=A2*B$2**. Copied down, A2 becomes A3, A4 ... but B$2 stays, because the price is only in B2. (With =A2*B2, C3 would use the empty B3 and show 0.)

B3 *fx* =SUM($B2:B2)

	A	B	C	D	E	F	G
1	No. of Overs	1	2	3	4	5	6
2	Runs per Over	13	14	10	7	15	11
3	Total	13	27	37	44	59	70

Runs of a cricket match *(Figure 7.37): B3 has* ***=SUM($B2:B2)****. Copied along row 3, it adds the runs from B2 to each over.*

7.21 The RANK function

	A
1	marks
2	85
3	74
4	95
5	14
6	56
7	96
8	32
9	17
10	84
11	41

RANK finds the position of a number in a list, for example a student's place in the class: **=RANK(number, ref, [order])**.

- **number** - the value to rank; **ref** - the range of the list
- **order** - 0 or left out: descending (the largest is 1st); 1: ascending (the smallest is 1st)

Formula	Order	Result
=RANK(A2,$A$2:$A$11)	descending (the largest value is 1st)	**3**
=RANK(A2,$A$2:$A$11,0)	descending (0 is the same as leaving it out)	**3**
=RANK(A2,$A$2:$A$11,1)	ascending (the smallest value is 1st)	**8**

Give the range as absolute references ($A$2:$A$11), so that it does not change when the formula is copied.

7.22 Sorting and filtering

Sorting puts data in **ascending** order (smallest to largest, A to Z) or **descending** order. Select a cell of the data, **Home tab** → **Sort & Filter**, and choose **Sort A to Z** or **Sort Z to A**. (**Data tab** → **Sort** sorts by several columns.)

1	**Student**	**Subject**	**Marks**
2	Pasindu	Sinhala	98
3	Kumar	Maths	92
4	Anas	Science	85
5	Sandasi	History	78

before sorting

1	**Student**	**Subject**	**Marks**
2	Sandasi	History	78
3	Anas	Science	85
4	Kumar	Maths	92
5	Pasindu	Sinhala	98

sorted by marks: smallest first

Filtering shows only the rows that meet a condition. The other rows are **hidden** for a while, not deleted:

1	Select any cell in the data range. The data must have a heading row.
2	Go to the **Data** tab. Click **Filter** in the **Sort & Filter** group.
3	Drop-down arrows appear in the heading cells. Click the arrow of a column.
4	Untick **Select All**, tick what you want (for example **Eastern**) and click **OK**.
5	To show all the rows again, click the arrow and select **Clear Filter**.

1	Employee	Region	Product	Sales (LKR)
6	Amir	Eastern	Laptop	150,000
8	Nawfa	Eastern	Keyboard	9,500

the sales table (textbook page 235) filtered: Region = Eastern

7.23 Charts

Numbers are hard to compare. A **chart** shows the data as a picture (Table 7.20):

Chart	Basic form	Purpose of the chart
Column chart		Comparing classes of data items in a group (group comparison).
Bar chart		Comparing classes of data items in a group (group comparison).
Line chart		Comparing classes of data items in a group (group comparison).
Pie chart		Comparing classes of data items as percentages.
XY scatter chart		Comparing data in pairs.

MS Excel 2010: Select the range of data (A1:F7), then **Insert** → **Chart** and select the type of chart. Chart title: **Layout** → **Chart Title** → **Above Chart**. Axis titles: **Layout** → **Axis Title**.

LibreOffice Calc: Select the range of data, then **Insert** → **Chart**, select the type and click **Finish**. Chart title and axis titles: **Insert** → **Title**.

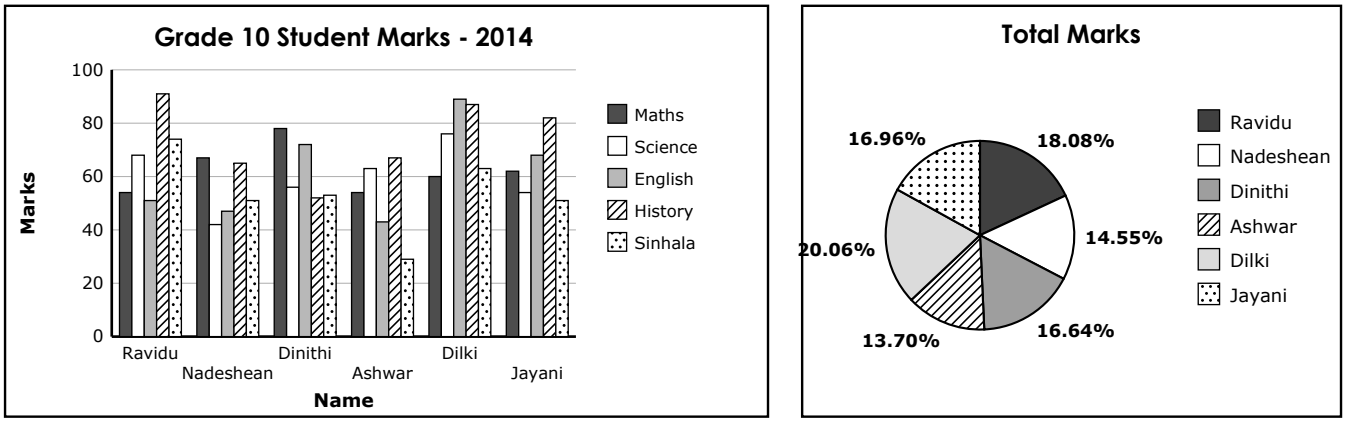


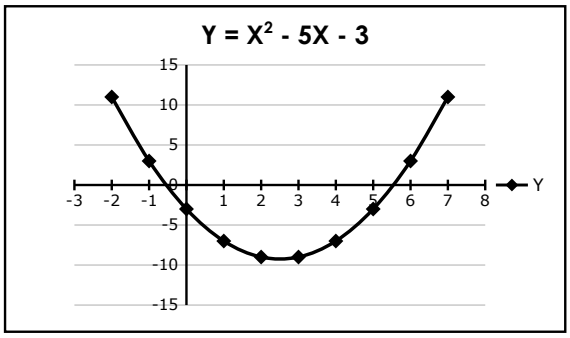
Figure 7.41: a column chart of the marks (A1:F7) with a chart title and axis titles, and a pie chart of the total marks (A1:A7 and G1:G7).

"Some things must stay fixed, like the $ in a formula. Keep your goals fixed too."

- THE KING OF ICT



Colour me!



An XY scatter chart. X is -2 to 7 in B1:K1. B2 has **=B1^2-5*B1-3**; copy it to K2. Select A1:K2, then **Insert → Chart → XY Scatter**.

The points are pairs (X, Y): (-2, 11), (-1, 3), (0, -3) ... (7, 11).

DO IT - PRACTICAL 4 (TEXTBOOK ACTIVITIES, PAGES 226 - 240)

Copying, ranking, sorting, filtering and charts. **Ask your teacher first.**

1	Type =A1+B1 in C1 and drag the fill handle down and across.
2	Make the pencils worksheet of 7.20 with =A2*B$2, then with =A2*B2. Compare.
3	Rank the marks of 7.21. Sort and filter the tables of 7.22.
4	Make Figure 7.41 and the charts of 7.23 with titles. Save your work.

A How does the formula change?

8 marks

Textbook Activity, pages 230 - 231. Each formula is in A1. Write the copied formula.

Formula in A1	Copied to	The new formula
=D2*$E2+G$2	B1	
=D2*$E2+G$2	C1	
=D2*$E2+G$2	A2	
=D2*$E2+G$2	A3	
=E2*F$2+$G2	B1	
=E2*F$2+$G2	A2	
=$G$2*$F2+G$2-H2	B1	
=$G$2*$F2+G$2-H2	A2	

B Relative or absolute?

6 marks

Tick the kind of each cell reference.

Reference	Relative	Row absolute	Column absolute	Row and column absolute
K3	☐	☐	☐	☐
H$2	☐	☐	☐	☐
$H2	☐	☐	☐	☐
$C$1	☐	☐	☐	☐
A$1	☐	☐	☐	☐
$A1	☐	☐	☐	☐

C Rank the marks

4 marks

Use the marks list A2:A11 of 7.21. Write the result.

1. **=RANK(A4,$A$2:$A$11)** .....
2. **=RANK(A10,$A$2:$A$11)** .....
3. **=RANK(A5,$A$2:$A$11,1)** .....
4. **=RANK(A8,$A$2:$A$11,1)** .....

D Sort and filter

6 marks

Use the marks of 7.21 and the sales table of 7.22.

1. The marks of 7.21 (A2:A11) are sorted from the smallest to the largest. What is in A2 now? .....
2. ... and what is in A11? .....
3. The sales table of 7.22 is filtered: Product = Laptop. Write the names that are shown. .....
4. ... and which row numbers are shown? .....
5. Now the filter is Region = Eastern. How many rows of data are shown? .....
6. Are the hidden rows deleted? Explain.

E Which chart?

4 marks

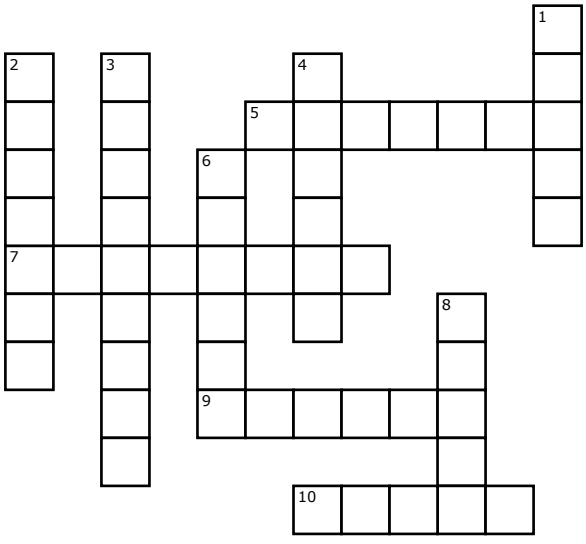
Write the best type of chart.

You want to ...	Chart
Show the percentage of the total marks that each student got.	
Draw $Y = X^2 - 5X - 3$ from pairs of X and Y values.	
Show the change in electric vehicles from 2010 to 2021.	
Compare the marks of six students in five subjects.	

F Crossword

10 marks

All the words are in this unit. Write one letter in each square.



ACROSS

5. It always begins with an equals sign (7)
7. A $ makes a cell reference ... (8)
9. The Excel 2010 toolbar with Home, Insert ... (6)
10. Text in a cell; it is left aligned (5)

DOWN

1. A pie or a column picture of the data (5)
2. The function that finds the mean of the values (7)
3. Cells made by columns and rows; it has a sheet tab (9)
4. The function that counts the non-empty cells (6)
6. It hides the rows you do not want to see (6)
8. B2:C4 is a ... of cells (5)

REVISION

Unit 7 in One Page

Read this the night before the test

Topic	The key facts
Spreadsheet	Rows and columns like a square ruled book. Excel (Microsoft), Numbers (Apple), LibreOffice Calc (The Document Foundation), OpenOffice Calc (Apache).
The window	Excel 2010: Quick Access Tool Bar, Ribbon and Tabs, Name Box, Insert Function, Formula Bar, Sheet Tabs, Status Bar. Calc: Menu Bar, Standard and Formatting Tool Bars.
Cells	Columns A ... Z, AA ... ; rows 1, 2, 3 ... Address = column letter + row number (D3). The active cell has a thick border.
Ranges	B2:B5 (along a column), A3:C3 (along a row), B2:C4 (6 cells). Ctrl selects more ranges.
Keys	Ctrl + Home = A1. Home = column A. Ctrl + End = the end of the data. Page Up / Page Down.
Data	Labels (left aligned), values (right aligned), formulas (begin with =).
Operators	+ - * / ^. First ( ), then ^, then * and /, then + and - (from left to right).
Functions	SUM, AVERAGE, MAX, MIN, COUNT (numbers), COUNTA (non-empty), ROUND(number, digits), IF(condition, value if true, value if false).
Formatting	Bold, Italic, Underline, alignment, decimal places. Format Cells: Number, Text, Percentage, Scientific, Date, Time, Currency.
Copying	Fill handle (drag) or Ctrl + C, Ctrl + V. Relative addresses change when copied.
Absolute	H$2 row absolute, $H2 column absolute, $H$2 row and column absolute.
RANK, sort	=RANK(A2,$A$2:$A$11): largest = 1st (order 0); 1 = ascending. Sort: ascending or descending.
Filter	Shows only the rows you want and hides the others (Data tab, Filter).
Charts	Column, bar, line: group comparison. Pie: percentages. XY scatter: data in pairs. Add a chart title and axis titles.

★ TICK WHEN YOU CAN DO IT

- ☐ I can name the parts of the Excel 2010 and LibreOffice Calc windows.
- ☐ I can write cell addresses and ranges of cells.
- ☐ I can enter labels, values and formulas.
- ☐ I can work out formulas in the right order of operators.
- ☐ I can use SUM, AVERAGE, MAX, MIN, COUNT, COUNTA, ROUND, IF and RANK.
- ☐ I can format cells and copy formulas with relative and absolute references.
- ☐ I can sort and filter data and create charts.

WORD POWER

Unit 7 Word List

Twenty-five words you must know

Word	What it means
Electronic spreadsheet	Software with rows and columns of cells for calculations, charts and more.
Worksheet	A sheet of cells made by columns and rows. Each worksheet has a sheet tab.
Column / Row	A vertical line of cells (A, B, C ...) / a horizontal line of cells (1, 2, 3 ...).
Cell	The box where a column and a row meet.
Active cell	The selected cell. It has a thick border.
Cell address	The column letter and then the row number of a cell, like D3.
Range of cells	A block of adjacent cells that is selected, like B2:C4.
Name Box	Shows the address of the active cell.
Formula Bar	Shows the formula or the contents of the active cell.
Ribbon / Tabs	The toolbar of Excel 2010 / the tabs that change the ribbon.
Menu Bar	In LibreOffice Calc: helps to select commands.
Label	Text: letters, numbers and special symbols. Left aligned.
Value	Data with a numerical figure, also dates, times and money. Right aligned.
Formula	Begins with = and uses values, cell addresses, operators and functions.
Operator	A sign for a calculation: + - * / ^
Precedence	The order of the operators: ( ), then ^, then * /, then + -.
Function	A ready-made formula such as SUM, AVERAGE or IF.
Format Cells	The dialog box to format numbers, text, dates, times and alignment.
Decimal places	The digits after the decimal point: 2.40 has two.
Fill handle	The small black square at the bottom-right corner of a selected cell.
Relative reference	A cell address like A1. It changes when the formula is copied.
Absolute reference	A cell address with $, like $A$1. It does not change when copied.
Sorting	Putting data in ascending or descending order.
Filtering	Showing only the rows that meet a condition. The others are hidden.
Chart	A picture of data: column, bar, line, pie or XY scatter.

"Learn the words today, and the formulas will follow tomorrow."

- THE KING OF ICT

END OF UNIT

Unit 7 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. Which company developed Excel?
   (a) Apple Inc (b) Microsoft Corporation
   (c) The Document Foundation (d) Apache Foundation
2. How many cells are there in the range C5:E7?
   (a) 3 (b) 6 (c) 9 (d) 12
3. What does =3+5^2 give?
   (a) 64 (b) 28 (c) 13 (d) 16
4. Which of these is a correct formula?
   (a) B5=A5*2 (b) =A5*2 (c) A5*2= (d) =A5x2
5. =COUNT(A1:A5) counts the cells that contain
   (a) labels (b) numbers (c) formulas (d) nothing
6. Which cell reference does not change when the formula is copied along a row or down a column?
   (a) H2 (b) H$2 (c) $H2 (d) $H$2
7. Which chart compares data items as percentages?
   (a) Column chart (b) Line chart (c) Pie chart (d) XY scatter chart
8. A - Labels are left aligned by default. B - A formula begins with =. C - Filtering deletes the hidden rows. Which are correct?
   (a) A, B only (b) B, C only (c) A, C only (d) A, B and C

2 Fill in the blanks

8 marks

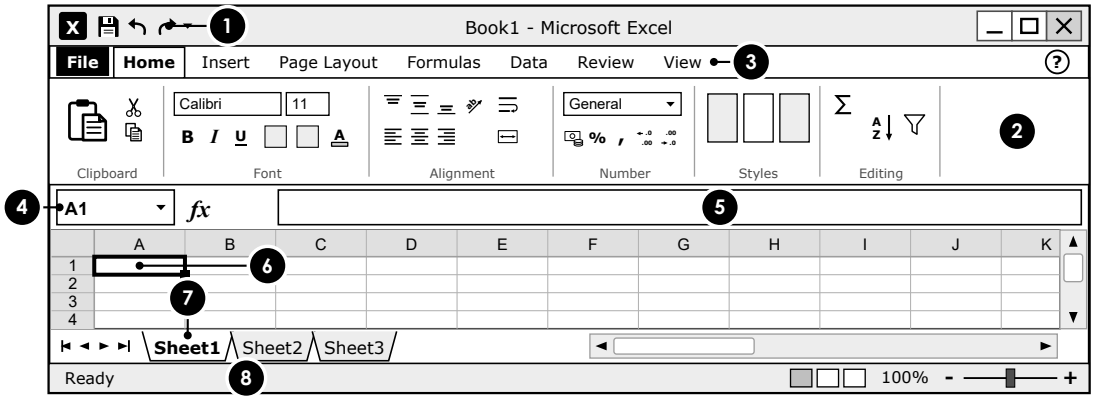
One word, number or sign in each blank.

1. In a cell address the column ..... comes first.
2. A block of selected adjacent cells is a ..... of cells.
3. The black square of a selected cell is the fill ..... .
4. The operator for multiplication is ..... .
5. =MAX(4,9,2) gives ..... .
6. =ROUND(7.25,1) gives ..... .
7. In LibreOffice Calc, Format Cells is in Format, then ..... .
8. The ..... sign makes a cell reference absolute.

3 Name the parts of the window

8 marks

This is the MS Excel 2010 window. Write the name of each numbered part.



1. ..... 2. ..... 3. ..... 4. .....

5. ..... 6. ..... 7. ..... 8. .....

4 Answer these

16 marks

Use the worksheet. Write the formulas in the space given.

	A	B	C	D	E	F	G	H
1	**Name**	**ICT**	**Maths**	**Science**	**Total**	**Average**	**Rank**	**Result**
2	Nimal	65	72	58				
3	Kamala	48	55	61				
4	Sahan	82	90	76				
5	Fathima	70	64	69				
6	Ruwan	39	45	50				
7								

1. A formula with a function for E2: the total marks of Nimal. (2 marks)

.....

2. A formula with a function for F2: the average marks of Nimal. (2 marks)

.....

3. A formula for B7: the highest ICT mark. (2 marks)

.....

4. A formula for H2: Pass if F2 is 50 or more, otherwise Fail. (2 marks)

.....

5. A formula for G2: the rank of Nimal by the total (it is copied to G6). (2 marks)

.....

6. What does =B2+C2*2 show? Show how you work it out. (2 marks)

.....

7. =A1*B$1 in C1 is copied to D2. Write the formula in D2. (2 marks)

.....

8. Which chart would you use to compare the five totals? Why? (2 marks)

.....